

Practice Questions for 2024-11-21-EconNetSimplex

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1 Questions

1. In the context of the network simplex method, what do the variables b_i and b_j typically represent?
 - A. Flows along the edges
 - B. Costs associated with the edges
 - C. Supply at node i and demand at node j
 - D. Slack variables
 - E. Dual variables
2. In the network simplex method, what does c_{ij} represent?
 - A. The flow from node i to node j
 - B. The supply at node i
 - C. The demand at node j
 - D. The cost of flow from node i to node j
 - E. The capacity of the edge between node i and node j
3. What is the significance of starting with a feasible solution on a spanning tree in the network simplex method?
 - A. It guarantees the optimal solution
 - B. It simplifies the problem by eliminating cycles
 - C. It ensures that all nodes are connected
 - D. It provides a basic feasible solution
 - E. It reduces the number of variables

2 Answers

1. In the context of the network simplex method, what do the variables b_i and b_j typically represent?
 - A. Answer A is incorrect because flows along the edges are represented by other variables, not b_i and b_j .
 - B. Answer B is incorrect because costs associated with the edges are represented by c_{ij} , not b_i and b_j .
 - C. Answer C is correct because b_i and b_j represent the supply at node i and the demand at node j , respectively, in the network flow problem.
 - D. Answer D is incorrect because slack variables are not directly represented by b_i and b_j in this context.
 - E. Answer E is incorrect because dual variables are not directly represented by b_i and b_j in this context.
2. In the network simplex method, what does c_{ij} represent?
 - A. Answer A is incorrect because the flow from node i to node j is represented by a different variable, not c_{ij} .
 - B. Answer B is incorrect because the supply at node i is represented by b_i , not c_{ij} .
 - C. Answer C is incorrect because the demand at node j is represented by b_j , not c_{ij} .
 - D. Answer D is correct because c_{ij} represents the cost associated with the flow from node i to node j .
 - E. Answer E is incorrect because the capacity of the edge is not directly represented by c_{ij} in this context.
3. What is the significance of starting with a feasible solution on a spanning tree in the network simplex method?
 - A. Answer A is incorrect because starting with a feasible solution on a spanning tree does not guarantee the optimal solution, but rather a starting point for the iterative process.
 - B. Answer B is incorrect because while a spanning tree does eliminate cycles, this is not the primary reason for starting with a feasible solution on a spanning tree.
 - C. Answer C is incorrect because while a spanning tree does connect all nodes, this is not the primary reason for starting with a feasible solution on a spanning tree.
 - D. Answer D is correct because a spanning tree provides a basic feasible solution, which is a starting point for the simplex method.
 - E. Answer E is incorrect because while a spanning tree does reduce the number of edges, it does not reduce the number of variables in the problem.